

Quiz 4

ECON 320 – Introduction to Mathematical Economics

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Part A – Multiple Choice Questions (1 point each)

1. The derivative of $\ln(x^2 + 1)$ is

- (a) $\frac{2x}{x^2 + 1}$
- (b) $2x \ln(x)$
- (c) $\frac{1}{x^2 + 1}$
- (d) $2(x^2 + 1)$

2. The derivative of e^{2x} is

- (a) e^{2x}
- (b) $2e^{2x}$
- (c) $2xe^{2x}$
- (d) e^{x^2}

3. If $y = e^{3x-5}$, then $\frac{dy}{dx} =$

- (a) e^{3x-5}
- (b) $3e^{3x-5}$
- (c) $5e^{3x-5}$
- (d) $e^{3x} - 5$

4. The natural logarithm function $\ln(x)$ is defined for

- (a) all real x
- (b) $x > 0$
- (c) $x < 0$
- (d) $x \neq 0$

5. If $f(x) = \ln(e^{2x})$, then $f'(x) =$

- (a) $2e^{2x}$
- (b) $2x$
- (c) 2
- (d) e^{2x}

6. The elasticity of $y = ae^{bx}$ with respect to x equals

- (a) bx
- (b) b
- (c) e^{bx}
- (d) $\frac{b}{x}$

7. The function $y = \ln(x)$ is

- (a) increasing and concave
- (b) decreasing and convex
- (c) increasing and convex
- (d) decreasing and concave

Part B – Analytical Problems (13 points total)

Question 1 (4 points)

The utility of income x is $U(x) = \ln(x + 2)$.

1. Compute $U'(x)$ and $U''(x)$.
2. Show that utility is increasing but concave.
3. Interpret the economic meaning of concavity in this context.

Question 2 (4 points)

A firm's production function is given by $Q(L) = 20L^{0.5}e^{-0.05L}$.

1. Derive the first-order condition for maximizing output.
2. Find the optimal labor input L^* .
3. Verify that it corresponds to a maximum.
4. Interpret the exponential term economically.

Question 3 (5 points)

The demand for a product is $Q(P) = 100e^{-0.2P}$.

1. Write the revenue function $R(P) = PQ(P)$.
2. Find the price P^* that maximizes revenue.
3. Compute the corresponding maximum revenue $R(P^*)$.
4. Verify the second-order condition.
5. Explain the trade-off between price and quantity in exponential demand.